

One product, in one sentence

Two fixed nodes watch a tabletop model. The laptop warns from rules and machine learning. Then an officer can drive a rover to look — because a warning appeared. Readings show disturbance. They do not prove a collapse.



Flowchart — two radios on purpose



Hardware we built — four boards



Software we built — one local website

Laptop stack

- Python + Flask · no login, no cloud
- SQLite history + CSV export
- Pages: /monitoring and /rover
- Simulate = labelled rehearsal
- Live USB never silently fakes data
- If USB drops → UNKNOWN, not green
- Run: python app.py --mode simulate
- Live: python app.py --mode live --serial auto

What the officer sees

Monitoring + AI

Node cards, WATCH / ALERT reason, Isolation Forest score, 30 s chart with MAE. No drive buttons here.

Rover Inspection

Hold FWD / REV / LEFT / RIGHT. Release = STOP. Distance, IR flag, MQ-7 raw. Open this page after a warning.

Machine learning — core product, on the laptop

Isolation Forest + LOF

Is this unusual vs this table's quiet normal?

- Features: tilt change, vibration, rate, Node A mm
- Shows score, top feature, train count
- Cannot clear a rule ALERT

Joint A vs B forest

Did one node move, or the whole model?

- Uses |A-B| tilt and vibration
- QUIET / LOCAL_A / LOCAL_B / COMMON
- Not a geology model

30 s sensor forecast

The hackathon "prediction"

- Next 30 s of tilt / vibration / crack mm
- Holdout MAE picks Ridge / Huber / linear
- Label: not a collapse prediction

Rules that latch (tabletop numbers, not mine standards)

NORMAL

Fresh data, below the watch line

WATCH

3° tilt or 2 mm gap, 3 samples

ALERT

6° or 4 mm, 3 samples — stays on

UNKNOWN

No packet for 5 s, or not calibrated

Ack = I saw it. Clear needs 3 NORMAL samples. Isolation Forest cannot turn ALERT off.

Simple honesty

We do say

- ☑ NORMAL / WATCH / ALERT / UNKNOWN on this rig
- ☑ Unusual vs this tabletop's normal + MAE
- ☑ 30 s forecast of the same signals
- ☑ MQ-7 raw ADC · IR is a near flag
- ☑ Remote drive · operator inspects

We do not say

- ☑ "The mine is safe / certified"
- ☑ Collapse or subsidence probability
- ☑ "Roof fails in N minutes"
- ☑ CO ppm · auto-brake · lidar
- ☑ Autonomous rover · camera AI · LoRa

Spoken close

Isolation Forest flags unusual combinations versus this rig's learned normal. A joint forest watches A vs B. Holdout-selected regression forecasts the next 30 seconds of those same signals with MAE. Rules remain the independent early-warning latch. The rover is remote inspection, not autonomy.